

- Q.28 Write a short note on ventilators.
- Q.29 Write a short note on computer applications in biomedical devices.
- Q.30 Write a short note on Pace maker.
- Q.31 Discuss NIBP.
- Q.32 Explain Pulse oxymeter.
- Q.33 Explain EEG machine with its diagram.
- Q.34 Discuss different types of bio electrodes.
- Q.35 Explain resting potential of a cell.

SECTION-D

- Note:** Long answer type questions. Attempt any two questions out of three questions. (2x10=20)
- Q.36 Draw and explain man-instrumentation system and its components.
- Q.37 Explain CAT scan with suitable diagram.
- Q.38 a) Describe defibrillators and its types
b) Discuss Glucometer

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6th Sem / Instrumentation & Control Engineering, EI Sub : Bio-Medical Instrumentation

Time : 3Hrs.

M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

- Q.1 Signal to noise ratio should be _____.
a) High b) Low
c) Both of these d) None of these
- Q.2 Value of action potential is
a) -70 mv b) -100 mv
c) +20 mv d) +10 mv
- Q.3 Body fluid contains charged atoms known as ions, these ions are _____.
a) Sodium b) Potassium
c) Chloride d) All of these
- Q.4 Electrodes are consists of _____.
a) German silver b) Stainless steel
c) Plastic d) All of above
- Q.5 PQRS complex is associated with _____.
a) Brain b) Heart
c) Muscle d) None of these

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- Q.6 Glucometer is used to measure _____.
 a) Blood sugar level b) Heat rate
 c) Temperature d) None of these
- Q.7 MICU expands as _____.
 a) Medical Intensive Care Unit
 b) Micro Intensive Care Unit
 c) Medical Interface Care Unit
 d) Medical Instrument Care Unit
- Q.8 Cardiovascular system is associated with _____.
 a) ECG b) EEG
 c) EMG d) None of these
- Q.9 _____ Instrument is known as diagnostic instrument
 a) Pulse oxymeter b) De-fibrillaters
 c) Pacemaker d) None of these
- Q.10 A machine that records electrical activity of muscles is known as.
 a) EEG b) ECG
 c) EMG d) None of these

SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

Q.11 Expand ICU.

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- Q.12 Cell is the smallest unit. (T/F)
- Q.13 Define transducer.
- Q.14 EEG machine is used to detect electrical activity of brain. (T/F)
- Q.15 Recorder is used to record the data. (T/F)
- Q.16 Blood pressure is measured by sphygmomanometer. (T/F)
- Q.17 Spiro meter's record is known as spirogram. (T/F)
- Q.18 Expand CRT.
- Q.19 Define Telemetry.
- Q.20 Stethoscope is a device which provides amplified sound energy. (T/F)

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

- Q.21 Differentiate between in-vivo and in-vitro measurement.
- Q.22 Discuss about respiration system of a human body.
- Q.23 Write a short note on bio electric potential.
- Q.24 Discuss EMG machine with its block diagram.
- Q.25 Write a short note on speech audio meter.
- Q.26 Discuss about bio-telemetry.
- Q.27 Discuss intensive care unit.

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